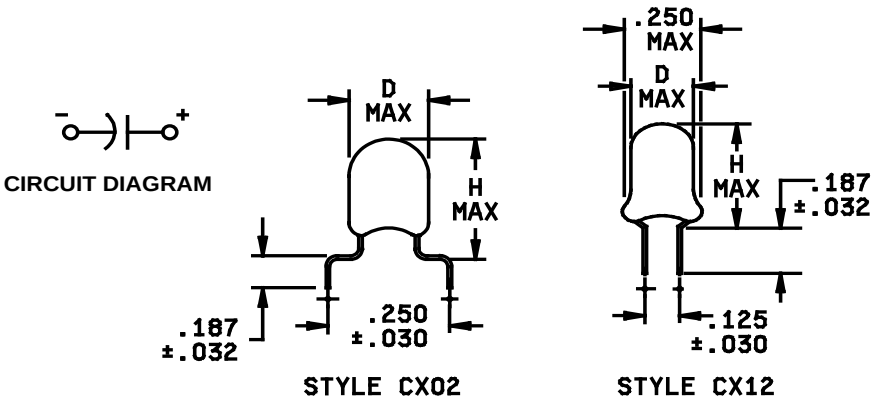
INCH-POUND
MIL-PRF-49137/2D
19 February 2009
SUPERSEDING
MIL-PRF-49137/2C
22 April 2004

PERFORMANCE SPECIFICATION SHEET

CAPACITORS, FIXED, ELECTROLYTIC (SOLID ELECTROLYTE), TANTALUM, POLAR, CONFORMAL COATED, NONHERMETICALLY SEALED, STYLES CX02 AND CX12

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-49137.



	Case size				Lead size	
	CX02					
Case size	Diameter (D)		Height (H)			
	Inches	mm	Inches	mm	Dia	AWG
A	0.175	4.45	0.425	10.80	0.020	24,22
B	0.250	6.35	0.500	12.70	0.020	24,22
C	0.350	8.89	0.650	16.51	0.025	22
D	0.400	10.16	0.750	19.05	0.025	22
	CX12					
E	0.175	4.45	0.350	8.89	0.020	24,22
F	0.250	6.35	0.500	12.70	0.020	24,22

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.
4. Lead spacing shall be measured within 0.050” (1.27 mm) from the point of emergence from the body or the bottom of the crimp.
5. Case size dimensions are maximum.

FIGURE 1. Dimensions and configuration.

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TABLE I. Style CX02 and CX12 characteristics.

Type designation 1/	DC rated voltage (85°C)	Capacitance (nominal)	DC leakage (max) +25°C	Dissipation Factor (DF)	Case size
	Volts	µF	µA	%	
CX-2D685-	6	6.8	1.0	6	A,E
CX-2D476-	6	47.0	3.0	8	B,F
CX-2D686-	6	68.0	4.5	8	B,F
CX02D157-	6	150.0	10.0	10	C
CX02D337-	6	330.0	20.0	10	D
CX-2F475-	10	4.7	1.0	6	A,E
CX-2F336-	10	33.0	3.5	8	B,F
CX02F107-	10	100.0	10.0	10	C
CX02F227-	10	220.0	20.0	10	D
CX-2H335-	15	3.3	1.0	6	A,E
CX-2H226-	15	22.0	3.5	8	B,F
CX02H686-	15	68.0	10.0	8	C
CX02H157-	15	150.0	20.0	10	D
CX-2J225-	20	2.2	1.0	6	A,E
CX-2J156-	20	15.0	3.0	8	B,F
CX02J476-	20	47.0	10.0	8	C
CX02J107-	20	100.0	20.0	10	D
CX-2K155-	25	1.5	1.0	6	A,E
CX-2K106-	25	10.0	2.5	8	B,F
CX02K336-	25	33.0	9.0	8	C
CX02K686-	25	68.0	15.0	8	D
CX-2M685-	35	6.8	2.5	6	B,F
CX02M226-	35	22.0	8.0	8	C
CX02M336-	35	33.0	10.0	8	D
CX02M476-	35	47.0	10.0	8	D
CX-2N104-	50	.1	1.0	4	A,E
CX-2N154-	50	.15	1.0	4	A.E
CX-2N224-	50	.22	1.0	4	A.E
CX-2N334-	50	.33	1.0	4	A.E
CX-2N474-	50	.47	1.0	4	A.E
CX-2N684-	50	.68	1.0	4	A.E
CX-2N105-	50	1.0	1.0	4	A,E
CX-2N155-	50	1.5	1.0	6	B,F
CX-2N225-	50	2.2	1.5	6	B,F
CX-2N335-	50	3.3	2.0	6	B,F
CX-2N475-	50	4.7	2.5	6	B,F
CX02N685-	50	6.8	3.5	6	C
CX02N106-	50	10.0	5.0	8	C
CX02N156-	50	15.0	6.0	8	C
CX02N226-	50	22.0	10.0	8	D

1/ Complete type designation will include additional symbols indicating style and capacitance tolerance.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Case material: Dipped, epoxy or plastic.

Leads: Solder-coated metal.

DC rated voltage: See table I.

Operating temperature: -55°C to +85°C.

DC leakage (DCL): See table I.

Capacitance (cap.): See table I.
Cap. tolerance: ±10% (K), or ±20% (M).

Dissipation factor (DF): See table I.

Stability at low and high temperature: In accordance with MIL-PRF-49137.

Step 1 (+25°):	DCL	-	See table I.
	Cap.	-	Within tolerance specified in table I.
	DF	-	See table I.
Step 2 (-55°C):	ΔCap.	-	Within -15 percent of step 1 measured value.
	DF	-	200 percent of table I.
Step 3 (+25°C):	DCL	-	See table I.
	ΔCap.	-	Within 10 percent of step 1 measured value.
	DF	-	See table I.
Step 4 (+85°C):	DCL	-	10 times 25°C limit (see table I).
	ΔCap.	-	Within +15 percent of step 1 measured value.
	DF	-	150 percent of table I.
Step 5 (+25°C):	DCL	-	See table I.
	ΔCap.	-	Within 10 percent of step 1 measured value.
	DF	-	See table I.

Surge voltage: In accordance with MIL-PRF-49137.

DCL	-	See table I.
ΔCap.	-	Within 10 percent of initial value.
DF	-	See table I.

Life: Method 108 of MIL-STD-202.
1,000 hours:

At 25°C:	DCL	-	200 percent of table I.
	ΔCap.	-	Within 10 percent of initial measured value.
	DF	-	See table I.

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These capacitors are intended to be used only where supplemental moisture protection is provided or for noncritical applications where hermetic moisture protection is not required.

NOTES:

Reference documents. In addition to MIL-PRF-49137, this specification sheet references the following documents.

MIL-STD-202

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094).

Changes from previous issues. The margins of this specification are marked with asterisks to indicate where changes from previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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